

Economics G6410 - Mathematical Methods for Economists - Fall 2019
Math Camp Exam

Department of Economics

Columbia University

Date: 6:10 - 8:00 p.m., September 3rd, 2019

Instructions:

This is a closed-book, closed-note exam, and no calculator is allowed. There are 10 questions in total, and the full grade is 100 points.

1. **(10 pts)** Let (X, d) be a metric space. Let $S \subset X$ be a finite set. Prove that S is closed, bounded, and compact.
2. **(10 pts)** Let $f : C \rightarrow \mathbb{R}$ be a continuous *bounded* function whose domain C is a *closed* subset of $\mathbb{R}^n$. Does f necessarily achieve the supremum of its values?
3. **(10 pts)** Consider the following matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

Is A diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If so, diagonalize it, i.e., find a 2×2 invertible matrix P s.t. $P^{-1}AP = \Lambda$, where Λ is a diagonal matrix.

4. **(10 pts)** Approximate $f(x) = e^{x+1}$ at $x_0 = 0$ with a Taylor polynomial of order three.
5. **(10 pts)** Prove that any collection of vectors that includes the zero-vector cannot be linearly independent.
6. **(10 pts)** Prove the following statements:
 - (a) In $(\mathbb{R}, d_2)$, let $\{x_n\}_{n=1}^{\infty}$ be a convergent sequence with limit x , and let b be a number such that $x_n \leq b$ for all n . Then $x \leq b$.
 - (b) If $x \in \text{int}S$ and $\{x_n\}_{n=1}^{\infty}$ is a sequence converging to x , then $x_n \in S$ for all sufficiently large n .
7. **(10 pts)** For each of the following subsets of $(\mathbb{R}^2, d_2)$ (plane), draw the set, state whether it is open, closed, neither, or both, and justify your answer in a word or two:
 - (a) $\{(x, y) : -1 < x < 1, y = 0\}$
 - (b) $\{(x, y) : x \text{ and } y \text{ are integers}\}$
 - (c) $\{(x, y) : x + y = 1\}$
 - (d) $\{(x, y) : x + y < 1\}$
 - (e) $\{(x, y) : x = 0 \text{ or } y = 0\}$
 - (f) $\{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$
8. **(10 pts)** Let $(\mathbb{R}^n, d_2)$ be a metric space. Prove that the intersection of compact sets is compact and that the finite union of compact sets is compact. Show that the infinite union of compact sets need not be compact.

9. **(10 pts)** Determine whether the following statement is true or false. If it is true, prove it. If not, provide a counterexample: Let (X, d_X) and (Y, d_Y) be metric spaces, and function $f : X \rightarrow Y$. The function f is a continuous function iff $f(E)$ is open in (Y, d_Y) for any set E open in (X, d_X) .
10. **(10 pts)** Consider the following problem:

$$\max_{(x,y) \in \mathbb{R}^2} x^2 + 4y^2$$

subject to

$$2x + 2y \leq 1 \tag{1}$$

$$x \geq 0 \tag{2}$$

$$y \geq 0 \tag{3}$$

- (a) Does a solution exist?
- (b) Should constraint (1) be binding at any solution?
- (c) Solve the program.